

**EVALUATION OF GROUND WATER QUALITY
AROUND IOL CHEMICALS AND PHARMACEUTICALS
LTD, BARNALA(PUNJAB)**

*A
Dissertation*

Submitted in partial fulfillment of the requirement for the
award of the degree of

MASTER OF
SCIENCE IN
ENVIRONMENTAL SCIENCE

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I hereby declare that the dissertation entitled “**Evaluation of Ground Water Quality Around IOL Chemicals and Pharmaceuticals Ltd., Barnala (Punjab)**” being submitted in partial fulfillment of the requirement for the award of the degree of Master of Science in Environmental Science, 2026, to the Gurukula Kangri(Deemed to be University), Haridwar (Uttarakhand), is my original piece of work and not submitted for the award of any degree fellowship or other title or prize.

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I am immensely pleased to forward the M.Sc. Dissertation of Mr. Prashant Kumar S/o Sh. Rajkumar on the topic entitled “**Evaluation of Ground Water Quality Around IOL Chemicals and Pharmaceuticals Ltd., Barnala (Punjab).**” for partial fulfilment of the degree of Master of Science in Environmental Sciences of Gurukul Kangri (Deemed to be University), Haridwar. The research work presented embodies the original work of the candidate and himself solely responsible for copyright and plagiarism.

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Content

Chapter No.	Title	Page No.
1.	INTRODUCTION	1-4
2.	REVIEW OF LITERATURE	5-8
3.	AIM AND OBJECTIVES	9-10
4.	MATERIALS AND METHODS	11-19
5.	RESULTS	20-28
6.	DISCUSSION	29-31
7.	SUMMARY	32-33
8.	REFERENCES	34-38

LIST OF SYMBOLS AND ABBREVIATIONS

S.No	Symbols	Name of	S.No	Symbols	Name of
1	%	Percentage	17	Fe	Iron
2	°C	Degree	18	gm	Gram
3	μS/cm	Micro-	19	K ⁺	Potassium
3	Abs	Absorbance	20	L	Liter
4	BIS	Bureau of	21	LPI	Leachate
5	Ca	Calcium	22	Mg	Magnesium
6	Cd	Cadmium	23	mg/L	Milligram per
7	Cl ⁻	Chloride	24	ml	Milliliter
8	Cl ⁻	Concentration	25	MSW	Municipal Solid
9	Conc.	Concentration	26	N	Normality
10	CPCB	Central	27	Na ⁺	Sodium
11	Cr	Chromium	28	Ni	Nickel
12	Cu	Copper	29	NO ₃ ⁻	Nitrate
13	EBT	ErioChrome	30	NTU	Nephelometric
14	EC	Electrical	31	P.A	Phenolphthalein
15	EDTA	Ethylene	32	Pb	Lead
16	FAS	Ferrous	33	pH	Potential of
17	Fe	Iron	34	PO ₄ ³⁻	Phosphate
18	g	Gram	35	SD	Standard
19	K ⁺	Potassium	36	SO ₄ ²⁻	Sulphate
20	L	Liter	37	TA	Total Alkalinity
21	LPI	Leachate	38	TDS	Total Dissolved
22	Mg	Magnesium	39	WHO	World Health
23	mg/L	Milligram per	40	WQI	Water Quality
24	ml	Milliliter	41	Zn	Zinc
25	N	Normality			

CHAPTER – 1

INTRODUCTION

Introduction

Water is a basic prerequisite for people and is moreover a imperative national asset and basic for IDSA (2010) states that moving forward the standard of living and financial status of society. As the stretch increases population and financial exercises both require water, so the water segment is where they intersect. will confront more and more trouble in closing the hole between supply and request. In general, groundwater has certain inborn important characteristics when compared to surface water. Changing these characteristics seem have results for the well-being of individuals (Lee et al., 2002). (Clark et al., 2003; Aiuppa et al., 2003). Moreover, around 97% of the earth's usable Groundwater, as characterized by (Delleur, 1999), serves as a common store for putting away freshwater, but ordinarily has a higher permanence time in the water cycle compared to surface waters that are more effortlessly accessible. Water is not essentially a product but a important asset that is essential. for the future era as well as for our current era (Rosu et al., 2014). It is one out of a group. the compounds that have a critical effect on life (Gorde and Jadhav, 2013).

Water is a constrained asset and is gradually getting to be a rare product in various locales over the globe. the same input dialect and with the same number of words: Identical dialect input, keeping up word number, and summarizing: "In a comparative manner" countries for get to water is fierce. agriculture, industry, and family utilization, all of which depend on water accessibility, constitute a restricting angle for the headway of the economy (Anitha et al., 2012).

Groundwater is an important natural asset and should therefore be safeguarded pollution caused by chemicals (EU, 2006). This is especially critical for underground water sources. interconnected ecosystems where surface water is greatly influenced by groundwater (Rozemeijer and Broers, 2007) and in cases where groundwater is utilized consumption of water in countries like the European Union and the United States (Solley et al., 1993). Low-income neighborhoods depend on untreated surface water and groundwater sources. Supplies for household and farming purposes are at highest risk from the effects of inadequate quality of water (Kazi et al., 2001). Regrettably, they are also the ones lacking sufficient infrastructure to regularly monitor water quality and enforce control. methods. (Ayoko et al., 2007). In today's global economic models, enduring socio-economic solutions are sought after. The sustainability of the resources plays a crucial role in the development of any community. water supplies (Lazhar et al., 2010). Sufficient amounts of suitable water are necessary. to address increasing demands from households, industries, and agriculture. The quality of groundwater is a crucial aspect. Highly delicate matter that goes beyond national borders. Many factors have an influence on it. Factors such as atmospheric chemistry, the underlying geology, and vegetation all play a role (natural decomposition of organic material) and human-caused factors. Groundwater is considered as one of the most valuable natural resources in the current study area because it provides the main source of drinking water for most people in the area. In the present day, the Strip is facing a water crisis due to groundwater contamination. a key issue in the past few years (Wang et al., 2018). Global water quality has significantly declined over recent years. Natural processes such as precipitation rate and weathering impact the environment over decades. Human activities such as deforestation, urbanization, and industrialization contribute to environmental degradation and soil erosion. exploitation of water

resources and release of municipal and industrial waste without treatment sewage (Chow et al., 2016). The quality of groundwater is at risk due to specific pollution sources like leaky. Oil storage tanks, sewage tanks, garbage dumps, etc., rather than focusing on groundwater exploration and dispersal origins of contamination. The primary danger to groundwater is diffuse agricultural pollution. The quality in various regions of the US was examined by (Nolan et al., 1997), Iran (Jalali and Kolahchi, 2008), Japan (Kumazawa, 2002) and is on the rise in developing nations such as India (Bijay-Singh et al., 1995). When it comes to human consumption, the primary source of demand is from urban areas. communities in need of water for drinking, sanitation, and drainage (UNESCO, 2004). Groundwater is typically of higher quality than surface water, unless it has been contaminated. can be gradually built near requirements, eliminating the necessity of large storage, processing, and distribution systems (Dapaah-Siakwan and Gyau, 2000). In an attempt to offer the populace with secure, sufficient, and lasting water supply, the crucial underground water reserves. have become more common in the sedimentary layers of the basin over time. taken advantage of Groundwater dating has effectively been utilized for identifying purposes.

Various studies, such as those by (Bohlke and Denver, 1995), have analyzed trends in the quality of groundwater. (Burow et al., 2008) the collection Monitoring data is collected from various monitoring wells throughout a groundwater system. linking all pollutant concentration measurements to the time of replenishment gives insight the foundation for identifying combined local patterns (Tesoriero et al., 2007). About 80% of diseases in India are thought to be caused by water, according to the World Health Organization. Health Organization stated that almost five million human fatalities transpire on a yearly basis. (Singh, 2004) stated that individuals are being affected by contaminated water every year. Human activities have a significant effect on the environment.

The activities have been so widespread that the self-cleaning abilities of the water bodies have diminished ability to a great extent (Sood et al., 2008). Supplying clean drinking water to rural areas Having a large number of people living in cities is required in order to avoid risks to public health (Lemo, 2002).

Before water is considered safe to drink, it must meet specific standards for its physical, chemical, and microbiological properties. microbiological criteria, aimed at guaranteeing the drinkability and safety of the water (Tebutt, 1983). Developing nations are experiencing alterations in groundwater resources. provide an additional source of drinkable water contamination of groundwater is one of the most troubling occurrences and the utilization of these waters for food concerns poses a threat to the well-being of individuals. Drinking water polluted with microorganisms can lead to outbreaks of disease (Marta et al., 2010). Because of the with population growth and economic development happening worldwide, there is a rising pressure on resources improve water quality management is necessary to achieve better outcomes. the endurance of systems for managing the quality of water. Thorough investigation into the associated subject It is very important to address obstacles and implement effective strategies to minimize their impact. Despite the implementation of the Water (Prevention and Control of Pollution) Act, issues persist. Just like in 1974, the water quality in India is still getting worse. Most of the water quality the rivers in India are monitored based on the guidelines provided by the Central Pollution Control Board (CPCB) requirements. Pollution in regions with plentiful water also leads to

water scarcity. sources of water. Groundwater is utilized as a safeguard against pollution. Pollutants from surface sources also penetrate into the environment without being spared. aquifer beneath, but as groundwater extraction rises, inorganic substances such as Fluoride and arsenic, which are found beneath the surface, also contaminate the extracted water. Groundwater pollution is extremely concerning because it accounts for 80 percent of the required domestic water supply formed by the source.

Industrial pollution, by its very nature, has the capability to create permanent damage reactions occurring in our surroundings are causing a significant danger to our survival (Saroj et al., 2016). Several studies have demonstrated the benefits of water. Pollution is negatively impacting human health by causing sickness and earlier death. The aforementioned text should be rewritten while maintaining the original language and word count. Environmental pollution and ecological degradation stemming from uncontrolled industrial growth started being a global concern in the 1970s, when the realization was made that Industry's mass production and society's mass consumption are causing depletion. resources and producing high volumes of solid waste and dangerous chemicals According to FEPA (1991), there has been a decline in the quality and quantity of groundwater resources decrease in the available water resources lowering the amount of water replenishing the aquifer, as well as growing the population. depleting the underground water reservoir has caused seawater intrusion in certain regions as a consequence in variations in pressure between the groundwater level and the level of sea water. H₂O in liquid form quality can be assessed by considering a combination of physical and chemical factors. Understanding the complexity of water quality data can be challenging due to the intricate factors involved. Therefore, This study aims to understand the complete and comprehensive picture of water quality. A new tool is required to classify the quality of domestic water in a simple way and format that is easy to comprehend. This tool offers an easy and cost-effective method for demonstrating discuss the decline in water quality observed in the current research, it is the perfect opportunity to interact with the media, global donors, and make decisions manufacturers. The primary goal of any evaluation is to determine the water quality status for specific purposes. research on monitoring the quality of water (Saeedi et al., 2009). The WQI, which stands for water quality index is a tool in mathematics that is utilized to convert extensive amounts of data on water quality. combine all the data into one number that indicates the level of water quality. Creating a WQI, in reality Engaging in zoning is a crucial step in the strategic organization of land utilization and water management. handling of a business. Quality of groundwater can be classified using a water quality index. research location and levels of pollution in the groundwater (Ramesh et al., 2010). The standard of quality typically has low levels of contaminants and balanced levels of essential minerals. The quality is poor due to low levels of pollutants such as nitrate and pesticides. In an ideal situation, the levels of pollutants in groundwater adhere to the standards set for drinking water, such as 45 milligrams per liter of nitrate in the BIS according to the regulations of 2001. Rising levels of contaminants or markers of pollution Pollutants are seen as a decline in the quality of groundwater.

CHAPTER-2

REVIEW OF LITERATURE

REVIEW OF LITERATURE

Binu Kumari et al., 2014 this study aims to comprehend the quality of groundwater by studying the physical and chemical properties of the Dhanbad town area in the state of Jharkhand. index for quality of water. The importance of different physicochemical factors like Total hardness, All dissolved solids (TDS), Chloride, Electrical conductivity, Salinity, Complete Total suspended solids (TSS), DO (Dissolved Oxygen), calcium hardness, magnesium Hardness, Alkalinity, and Sulphate were measured at 1700 mg/l, 966 mg/l, and 908.8 mg/l respectively. 1790 $\mu\text{S}/\text{cm}$ equals to 1455 mg/l, 6.9 is equal to 524 mg/l, 1207.5 mg/l is the same as 173 mg/l, 372 mg/l is equivalent to 0.674 mg/l in the first case, and mg/l in the second case. 75% of the water samples are classified in this category based on the results. 25% of the water sample is classified as category C (Poor water) while B (Good water) accounts for the rest. The main reason for this is the high levels of TDS and total hardness.

Research on the water quality index of groundwater in Aland taluka, Gulbarga (India) focuses on assessing the quality of well water for both rural and urban populations using WQI, considering groundwater characteristics and quality assessment as key components. Water samples from three locations in ten villages in Aland taluka were collected, following standard procedures, and analyzed for various components such as pH, total hardness, calcium, magnesium, chloride, total dissolved solids, iron, fluoride, nitrate, and sulfate. The calculation of water quality index used the BIS-10500-1991 standards (**SrinivasKushtagi and Padaki Srinivas, 2011**).

Rajankar et al., (2013) conducted an assessment of the quality of tube well water. W.Q.I. in Wardha, India. Some of the textile factories in Wardha district were evaluated using the Water Quality Index (W.Q.I.) assessed. Calculation is based on factors like pH, turbidity, temperature, dissolved oxygen, and biochemical oxygen demand in the area designated for residential, commercial, and agricultural purposes. Some different physico-chemical properties Parameters include E.C., Total hardness, calcium hardness (as CaCO_3), and SO_4 (sulphate). Chloride ions (Cl^-), sodium ions (Na^+), and potassium ions (K^+) are among the final parameters that are compared with the Bureau of Indian Standard, abbreviated as B.I.S.

Kiran and Mehta, (2010) conducted a study in which they collected groundwater samples from different locations in Kankrej taluka of Gujarat state, India. They analyzed various physico-chemical parameters such as temperature, color, odor, turbidity, EC, pH, TDS, total alkalinity, and

concentrations of ions including fluoride, chloride, sulfate, nitrate, calcium, magnesium, sodium, and potassium. The results were contrasted with the drinking water standards set by the Indian Standard (IS) and the World Health Organization (WHO). The current research has shown that the water samples studied met most of the physicochemical standards, but the Total Dissolved Solids (TDS) levels were found to be exceeding the limits in all samples. In order to prevent potential negative impacts, it is recommended by IS to treat water adequately before using it as drinking water. Hence, the public needs to be informed about the quality of drinking water. Managing valuable natural liquid resources is a crucial requirement in present-day society.

Gorai and Subhash Kumar, (2013) summarized the present groundwater quality status and examined the spatial distribution of groundwater quality in the Ranchi Municipal Corporation (RMC) area for risk evaluation. Determine the most suitable match for the twelve parameters related to water quality (calcium, magnesium, iron, nitrate, manganese, etc.). sodium, potassium, pH, TDS, total hardness, alkalinity, and turbidity are all factors that can be measured. The findings will help planners and decision makers create policies for the effective management of groundwater resources.

Water samples were taken from nine locations near a sugar factory within a 4 km radius and tested for physical properties. The water quality index for various groundwater sources in the industrial zone was determined, showing a range from good to very poor quality (**Ratnakant Sheth, 2011**).

Anita Pius and Charmaine Jerome, (2010) determined that groundwater is a crucial and indispensable part of our life support system. Analysis of specific physico-chemical characteristics of groundwater samples extracted from certain bore wells led to the calculation of the water quality index (WQI) using the collected data. The values of the WQI varied between 49.2 and 409.94. The Pearson correlation was conducted to determine the connection between WQI and quality of life. The correlation coefficient 'r' was found to be -0.499 with a significance level of $p < 0.001$.

Chandaluri Subba Rao, (2010) Conducted a physico-chemical analysis of well and bore well water samples collected from eight sampling sites in the Guntur rural area in February 2010. Determined water quality index (WQI) to evaluate if water is suitable for drinking. The results of WQI from various sampling stations ranged from 38.3 to 42.6.

Analysis was conducted on the physicochemical properties of groundwater (two open wells, two bore wells) and municipal water in Amalner town. A total of 15 parameters were examined and the findings were evaluated against the criteria set by the WHO. (**Patil and Patel, 2009**) discovered that a small number of sampling sites had contaminated underground water, while Dekhuroad had

physicochemical parameters that met water quality standards, indicating good-quality water suitable for drinking.

Omofonmwan, (2009) investigated how solid waste impacts the quality of underground water in Benin metropolis, Nigeria. The examination of raw water from eight wells near refuse dumps in the city found that the wastes create leachates and gases as they decompose and are washed by rainwater into groundwater through percolation and infiltration.

CHAPTER - 3

AIM AND OBJECTIVES

AIM AND OBJECTIVES

Aim:

This research focuses on providing a comprehensive picture of groundwater quality. The objectives of the research are:

Objectives

- Collect the groundwater samples from the three selected monitoring locations.
- To identify the groundwater quality by analysing the physico-chemical parameters..
- To assess the groundwater quality for drinking purposes by comparing the observed dataset with the Indian standards permissible limit proposed by BIS (2012).

CHAPTER - 4

MATERIALS AND METHODS

Materials and Methods

The Materials and Methods include the following aspects:

- Description of study area.
- Collection of samples.
- Analysis of Physicochemical parameters

Description of study area: The present study was carried out at three different sites demarcated near Iol Chemicals and Pharmaceuticals, Kacheri Chowk, and Railway Station, Barnala. The following sites were selected for the present study, as mentioned in Table 1:

S.No.	Sampling Sites	Location coordinates
1.	Near Iol chemicals and pharmaceuticals	30.9237°N, 75.8077° E
2.	Kacheri chowk	30.3603°N, 75.5328° E
3.	Railway Station Barnala	30.3710°N, 75.5498° E

SITE PROFILE-Site1: NEAR IOL CHEMICALS AND PHARMACEUTICALS

The first site chooses for Ground water sample Near Iol chemicals and pharmaceuticals. The Latitude and Longitude of the site is 30.9237°N, 75.8077° E



Image:1



Image:2

SITE2: KACHERI CHOWK

The second site of my study area is Kacheri chowk The Latitude and Longitude of the site are 30.3603°N, 75.5328° E



Image:3



Image:4

Site3: RAILWAY STATION BARNALA

The last site of my study area is Railway Station Barnala. The Latitude and Longitude of the site 30.3710°N, 75.5498° E



Image:5



Image:6

Collection of samples

Sampling and analysis have been done as per the standard methods prescribed in BIS, CPCB, APHA (2012) and Trivedi and Goel (1986). The water samples were collected from taps which were located at the above-mentioned three sites. For the collection Samples bottles were used.

Physicochemical parameters

pH, TDS (mg/l), Alkalinity (mg/l), Turbidity (mg/l), Total Hardness (mg/l), Calcium (mg/l), Magnesium (mg/l).

It was recorded with the help of good grade mercury field Celsius thermometer having the accuracy of 0.1 to 50.

Procedure

- For determination of Ground temperature water was collected in suitable container.
- After some time, collection of the sample was inserting mercury thermometer should be of small thermal capacity to attain equilibrium rapidly.
- While taking reading the scale of the thermometer should be immersed in the water up to the level of mercury in the capillary column.

pH: The pH is the concentration of hydrogen ions in a solution. A solution with a high concentration of hydrogen ions is acidic. A solution with a low amount of hydrogen ions is basic, or also known as alkaline.

PH Meter: PH Meter is an electronic instrument consisting of a special bulb that is sensitive to hydrogen ions that are present in the test solution. The signal produced by the bulb is amplified and sent to an electronic meter connected to the bulb, which measures and display the pH reading. It gives more precise values than the pH papers.

Total Dissolved Solids

Suspended solids are the portions of solids that are retained on a filter of standard specified size (generally 2.0 μ) under specific conditions. Water with high-suspended solids is unsuits factory for bathing, industrial and other purposes.

Principle

A well – mixed sample is filtered through a weighed standard glass fiber filter and the residue that is retained on the filter is dried to a constant weight at 103-105°C. The increase in the weight of the filter determines the total suspended solids.

Apparatus

Evaporating dishes, -100 ml porcelain dish, steam bath, drying oven, desiccators, Monopan balance and measuring jars. Procedure

- Wash Filter Paper.
- Dry evaporating dish and weigh
- Stir sample
- Pipette 50ml while stirring
- Filter and wash three times
- Transfer filtrate to evaporating dish and dry
- Cool and weigh
- Calculate in mg/l

Calculation

$$\text{Dissolved Solids mg /L} = (A - B) \times 1000/\text{ml sample}$$

Where;

$$A = \text{weight of dried residue + dish (mg)} \quad B = \text{weight of dish (mg)}$$

Alkalinity

Alkalinity is the water's capacity to resist changes in pH that would make the water more acidic. Alkalinity refers to the capability of water to neutralize acid. This is really an expression of buffering capacity. A buffer is a solution to which an acid can be added without changing the concentration of available H⁺ ions (without changing the pH) appreciably. It essentially absorbs the excess H⁺ ions and protects the water body from fluctuations in pH.

Principle

Most alkalinity in surface water comes from calcium carbonate (CaCO_3) that comes from rocks and soil. Limestone contains high level of calcium carbonate. The process is enhanced if the rocks and soil have already been broken up before entering the water. The dissolved minerals get into the water through construction and other processes.

Apparatus

Stand, Burette, funnel, conical flask, beaker etc.

Procedure

- Take 50 ml sample in a flask
- Add six drops of phenolphthalein indicator in the sample (water), note the initial reading of the burette containing H_2SO_4 ($N=0.02$)
- Start the titration till the color changes and note the reading of the burette,
- Calculate the phenolphthalein alkalinity using the formula $\text{alkalinity} = (\text{final reading} - \text{initial reading}) \times 100/50$.
- Now add six drops of brome cresol green in the mythyl solution which turns the color red to greenish one. Note: the initial reading of the burette and start the titration till the color changes to gray and note the final reading Calculate total Alkalinity by using the formula.

Calculation

Total alkalinity = (final reading - initial reading) x 100/50

Turbidity (Nephelometric method) Requirements

Nephelo turbidity meter with sample cells, volumetric flask and amber glass bottle etc.

Reagents

Solution I: Dissolve 1.0 gm hydrazine sulphate, $(\text{NH}_2)_2$, H_2SO_4 in distilled water and dilute to 100 ml in a volumetric flask.

Solution II: Dissolve 10 gm hexamethylenetetramine, $(\text{CH}_2)_6\text{N}_4$, in distilled water and dilute to 100 ml in a volumetric flask.

4000 NTU Suspension: In a flask mix 5.0 ml of solution I and 5 ml of solution II.

Let Stand for 24 hours at $25 \pm 0^\circ\text{C}$. This result in a 4000 NTU Suspension. Store in a glass bottle.

Dilute 4000 NTU stock solution with distilled water to prepare dilute standard just use and discard after use.

Method

Calibrate the Nephelo turbidity meter with the 100 NTU suspensions and verify the setting with a reading on the 10 NTU suspensions. Then thoroughly shake sample, wait until air bubbles disappear and pour sample into turbidity meter tube. Read turbidity directly from instrument display. In case the turbidity is going beyond the scale dilute the sample and measure the turbidity. The dilution factor should be taken into account for the final calculation of the result.

Calculation

Turbidity (NTU) = NTU measured/proportion of sample in dilution

Total Hardness EDTA Titrimetric Method Requirements-

Volumetric flask, conical flask, burette, pipette, measuring cylinder, burette stand, pipette stand and wash bottle etc. **Reagents**

- **Buffer Solution:** Dissolve 16.9 gm NH_4Cl in 143 ml conc. NH_4OH . Add 1.25 gm magnesium salt of ethylene diamine tetra acetic acid (EDTA) and dilute to 250 ml with distilled water. Store in a plastic bottle stopper tightly for no longer than one month.
- **Eriochrome Black T Solution (as Indicator):** Dissolve 0.5 gm dye in 100 ml triethanolamine or 2 ethylene glycol monomethyl ether. The salt can also be used in dry powder form by grinding 0.5 gm dye with 100 gm NaCl .
- **Standard EDTA Titrant (0.01M):** Weight 3.723 gm di-sodium salt of EDTA, dehydrate, dissolve in distilled water and dilute to 1000 ml. Store in polyethylene bottle.
- **Standard calcium solution:** Weight 1 gm anhydrous CaCO_3 in a 500 ml flask. Add 1 + 1 HCl slowly through a funnel till all CaCO_3 is dissolved. Add 200 ml distilled water and boil for a few minutes to expel CO_2 . Cool and add a few drops of methyl red indicator and adjusts to the intermediate orange color by adding 3N NH_4OH or 1 + 1 HCl , as required. Transfer quantitatively and dilute to 1000 ml with distilled water, 1 ml=1mg CaCO_3 .

Method

Standardize the EDTA titrant against standard calcium solution using procedure given below- Take 50 ml sample in a conical flask add 2 ml buffer solution and 2 drops EBT indicator solution, wine red color appears. Titrate with EDTA till the color changes to blue. Note the volume of EDTA used. Calculate the total hardness by following formula-

Formula

Total Hardness (EDTA) mg/l = $A \times B \times 1000 / \text{volume of sample}$

Where:

A = ml EDTA used, B = mg CaCO₃ equivalent to 1 ml EDTA

Ca & Mg HARDNESS Titrimetric method Requirements:

Volumetric flask, conical flask, burette, pipette, measuring cylinder, burette stand, pipette stand and wash bottle etc.

Reagents:

- **Buffer Solution:** Dissolve 16.9 gm NH₄Cl in 143 ml conc. NH₄OH. Add 1.25 gm magnesium salt of ethylene diamine tetra acetic acid (EDTA) and dilute to 250 ml with distilled water. Store in a plastic bottle stoppered tightly for no longer than one month.
- **Standard EDTA Titrant (0.01M):** weight 3.723 gm di-sodium salt of EDTA, dehydrate, dissolve in distilled water and dilute to 1000 ml. Store in polyethylene bottle.
- **Standard Calcium Solution:** weight 1 gm anhydrous CaCO₃ in a 500 ml flask. Add 1 + 1 HCl slowly through a funnel till all CaCO₃ is dissolved. Add 200 ml distilled water and boil for a few minutes to expel CO₂. Cool and add few drops of methyl red indicator and adjusts to the intermediate orange colour by adding 3N NH₄OH or 1 + 1 HCl, as required. Transfer quantitatively and dilute to 1000 ml with distilled water, 1 ml-1mg CaCO₃
- **Sodium hydroxide:** 4 gm NaOH dissolved in 100 ml distilled water.
- **Murexide (ammonium pupurate) Indicator:** 75 gm of the indicator is dissolved in 50 gm absolute ethylene glycol.

Method:

Take 50 ml sample in conical flask raise the pH to 12-13 by adding 2 ml NaOH. Add 1-2 drops of indicator and titrate with EDTA until solution becomes purple from pink.

Calculation

$$\text{Calcium hardness as CaCO}_3 \text{ (mg/l)} = A \times B \times 100 \times 1000 / \text{volume of sample}$$

Where:

$$A = \text{ml EDTA used, } B = \text{mg CaCO}_3 \text{ equivalent to 1 ml EDTA}$$

Magnesium hardness

Magnesium hardness is determined by subtraction of Ca hardness from the total hardness
 $\text{Magnesium hardness (mg/l)} = \text{total hardness} - \text{calcium hardness}$

Calcium

Calcium is determined by multiplying the Ca hardness by 0.4 i.e. calcium as Ca (mg/l) = calcium hardness x 0.40

Magnesium

Magnesium is determined by multiplying the magnesium hardness by 0.243 i.e. Magnesium as Mg (mg/l) = magnesium hardness x 0.243.

Method

Take 50 ml sample in a conical flask add 20 ml buffer solution and 2 ml conditioning reagent, mix with magnetic stirrer. While stirrer add a pinch of BaCl₂ crystal stir for 1 minute at constant speed, after stirring pour solution into cell and measure absorbance at 420 nm in spectrophotometer. Prepare standard from 5 mg to 40 mg SO₄²⁻/litre and measure the absorbance. Prepare standard graph between absorbance and concentration of sulphate ions.

CHAPTER - 5

RESULT

RESULT

SAMPLING SITE: 1 NEAR IOL CHEMICALS AND PHARMACEUTICALS

pH: During the present study the minimum pH was observed 7.8 in January whereas 8.2 in March was maximum value of pH. This difference may be due to the fact that carbon-dioxide dissolved in water to form carbonic acid. Although this is a weak acid, large amount of it will lower the pH and when water with low pH values come in contact with certain chemicals and metals, this often makes them more poisonous than normal.

Total Dissolved Solids (TDS): Throughout the present study the minimum TDS was observed 294 mg/L in January and maximum value was noticed 319 mg/L in February. Most often, high levels of TDS are caused by the presence of potassium, chlorides and sodium. These ions have little or no short-term effects, but toxic ions (lead arsenic, cadmium, nitrate and others) may also be dissolved in the water.

Alkalinity: In the present study the minimum Alkalinity was observed 128 mg/L in March and maximum value of Alkalinity was observed 142 mg/L in February. It may be due to the presence of large amount of carbonates and bicarbonates ions from rocks and other resources.

Turbidity: All through the present study, the minimum Turbidity was observed 0.4 mg/L in March and maximum Turbidity was noted as 0.8 mg/L in January. Turbidity in open water may be caused by growth of phytoplankton. Human activities that disturb land, such as construction, mining and agriculture, can lead to high sediment levels entering water bodies during rain storms due to storm water runoff.

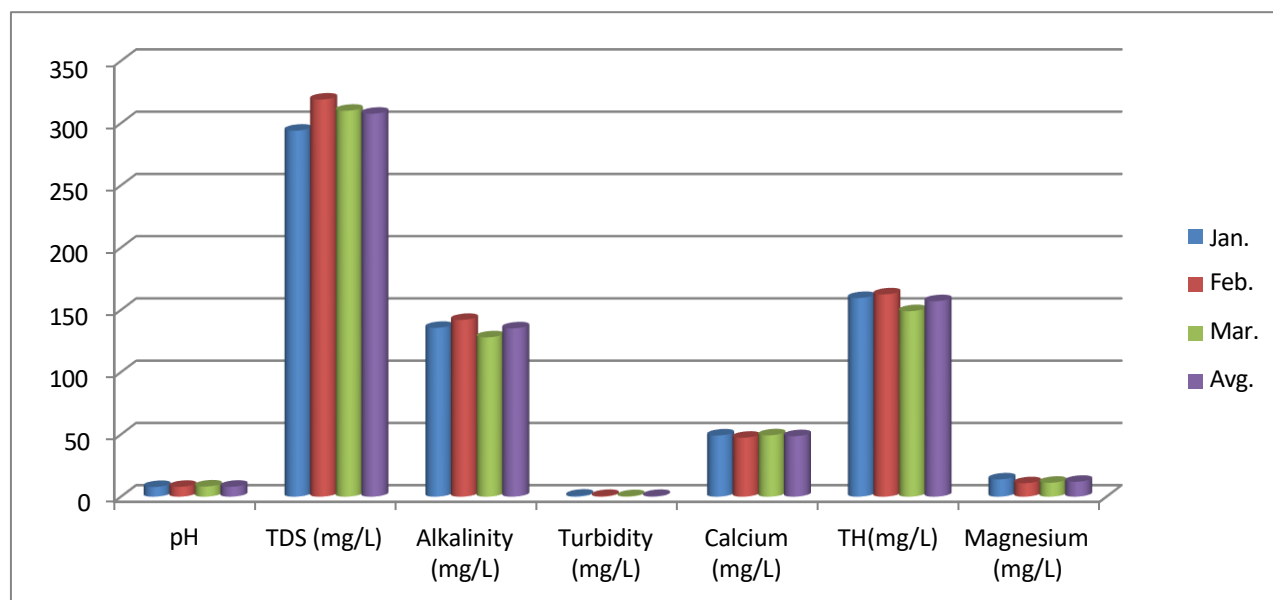
Total Hardness: Through the present study the minimum Total Hardness was observed 149 mg/L in March and maximum value was noticed 162.5 mg/L in February. This difference in hardness may arise due to the presence of Sulphate, carbonates, silicates arising from various industrial effluent and natural sources like rocks. Hardness is the property of water which prevents the lather formation with soap and increases the boiling points of water.

Calcium: In the present study the minimum Calcium was observed 47.46 mg/L in February and maximum value was observed 47.29 mg/L in March. The Calcium is naturally present in water. It may dissolve from rocks such as limestone, marble, calcite, dolomite, gypsum, fluorite and apatite. Calcium is a determinant of water hardness, because it can be found in water as Ca^{2+} ions.

Table:2. Physico-chemical parameters analyzed at sampling site 1

Months/ Parameters	Jan.	Feb.	Mar.	Avg.	BIS Standard for DW (Permissible Limit)
pH	7.8	7.92	8.2	7.97	6.5-8.5
TDS(mg/L)	294	319	310	307.6	2000
Alkalinity(mg/L)	135.61	142	128	135.20	600
Turbidity(NTU)	0.8	0.6	0.4	0.6	5
Calcium(mg/L)	49.17	47.36	49.29	48.60	200
TH(mg/L)	159.6	162.5	149	157.03	600
Magnesium(mg/L)	13.9	10.8	11.2	11.96	100

Graph1: Showing the parameters at sampling sites 1.



Magnesium: In the course of the present study the minimum Magnesium was determined 10.8 mg/L in February whereas 13.9 mg/L was maximum value of Magnesium in January. Magnesium is washed from rocks and subsequently ends up in water. Magnesium has many different purposes and consequently may end up in water in many different ways. Chemical industries add magnesium to plastics and other materials as a fire protection measure or as filler. It also ends up in the environment from fertilizer application and from cattle feed. Magnesium Sulphate is applied in beer breweries, and magnesium hydroxide is applied as a flocculent in wastewater treatment plants.

SAMPLINGSITE:2- KACHERI CHOWK

pH: In the present study the minimum pH was observed 7.12 in January whereas 7.28 in February was maximum value of pH. As mentioned above this difference may be due to the fact that carbon-dioxide dissolved in water to form carbonic acid. Although this is a weak acid, large amount of it will lower the pH and when water with low pH values come in contact with certain chemicals and metals, this often makes them more poisonous than normal.

TDS: In the course of the present study the minimum TDS was observed 260 mg/L in March and maximum value was noticed 281 mg/L in February. Most often, high levels of TDS are caused by the presence of potassium, chlorides and sodium. These ions have little or no short-term effects, but toxic ions (lead arsenic, cadmium, nitrate and others) may also be dissolved in the water

Alkalinity: Throughout the present study the minimum Alkalinity was observed 104 mg/L in March and maximum value of Alkalinity was 110 mg/L in January. It may be due to the presence of large number of carbonates and bicarbonates ions from rocks and other resources.

Turbidity: In the present study, the minimum Turbidity was observed 0.2NTU in February and maximum Turbidity was noted as 0.3 NTU in January & March. Turbidity in open water may be caused by the growth of phytoplankton. Human activities that disturb land, such as construction, mining and agriculture, can lead to high sediment levels entering water bodies during rain storms due to storm water runoff.

Total Hardness: In the present study the minimum Total Hardness was observed 147 mg/L in January and maximum value was noticed 153 mg/L in March. This difference in hardness may arise due to the presence of Sulphate, carbonates, silicates arising from various industrial effluent and natural sources like rocks. Hardness is the property of water which prevents the lather formation with soap and increases the boiling points of water.

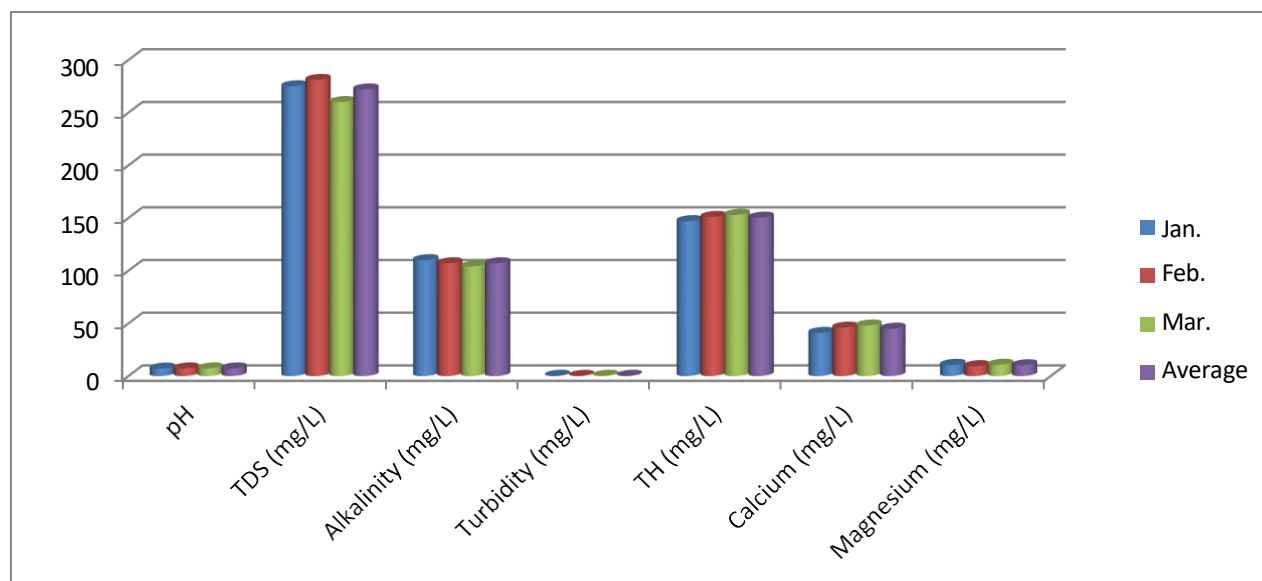
Calcium: Throughout the present study the minimum Calcium was observed 41 mg/L in January and maximum value was observed 48 mg/L in March. The Calcium is naturally present in water. It may dissolve from rocks such as limestone, marble, calcite, dolomite, gypsum, fluorite and apatite. Calcium is a determinant of water hardness, because it can be found in water as Ca^{2+} ions.

Magnesium: In the course of the present study the minimum Magnesium was determined 9.4 mg/L in February whereas 10.7 mg/L was maximum value of Magnesium in March. Magnesium has many different purposes and consequently may end up in water in many different ways. Chemical industries add magnesium to plastics and other materials as a fire protection measure or as filler. It also ends up in the environment from fertilizer application and from cattle feed. Magnesium Sulphate is applied in beer breweries, and magnesium hydroxide is applied as a flocculant in wastewater treatment plants.

Table:3. Physicochemical parameters analyzed at sampling site-2

Months/P arameters	Jan.	Feb.	Mar.	Average	BIS Standard for DW (Permissible Limit)
pH	7.12	7.28	7.29	7.23	6.5-8.5
TDS(mg/L)	275	281	260	272	2000
Alkalinity(mg/L)	110	107	104	107	600
Turbidity(NTU)	0.3	0.2	0.3	0.26	5
TH(mg/L)	147	151	153	150.33	600
Calcium(mg/L)	41	46	48	45	200
Magnesium(mg/L)	10.6	9.4	10.7	10.23	100

Graph2: Showing the parameters at sampling sites 2.



SAMPLINGSITE:3 RAILWAY STATION BARNALA

pH: Throughout the present study the minimum pH was observed 7.02 in January whereas 7.21 in February was maximum value of pH. As mentioned above this difference may be due to the fact that carbon-dioxide dissolved in water to form carbonic acid. Although this is a weak acid, large amount of it will lower the pH and when water with low pH values come in contact with certain chemicals and metals, this often makes them more poisonous than normal.

TDS: The minimum TDS during the present study was observed 280 mg/L in January and maximum value was noticed 310 mg/L in February. Most often, high levels of TDS are caused by the presence of potassium, chlorides and sodium. These ions have little or no short-term effects, but toxic ions (lead arsenic, cadmium, nitrate and others) may also be dissolved in the water.

Alkalinity: In the present study the minimum Alkalinity was observed 118 mg/L in January and maximum value of Alkalinity was 128 mg/L in February. It may be due to the presence of large number of carbonates and bicarbonates ions from rocks and other resources.

Turbidity: In the course of the present study, the minimum Turbidity was observed 0.4 NTU in February and maximum Turbidity was noted as 0.5 NTU in January & March. Turbidity in open water may be caused by growth of phytoplankton. Human activities that disturb land, such as construction, mining and agriculture, can lead to high sediment levels entering water bodies during rain storms due to storm water runoff.

Total Hardness: In the present study the minimum Total Hardness was observed 142 mg/l in February and maximum value was noticed 156 mg/L in March. This difference in hardness may arise due to the presence of Sulphate, carbonates, silicates arising from various industrial effluent and natural sources like rocks. Hardness is the property of water which prevents the lather formation with soap and increases the boiling points of water.

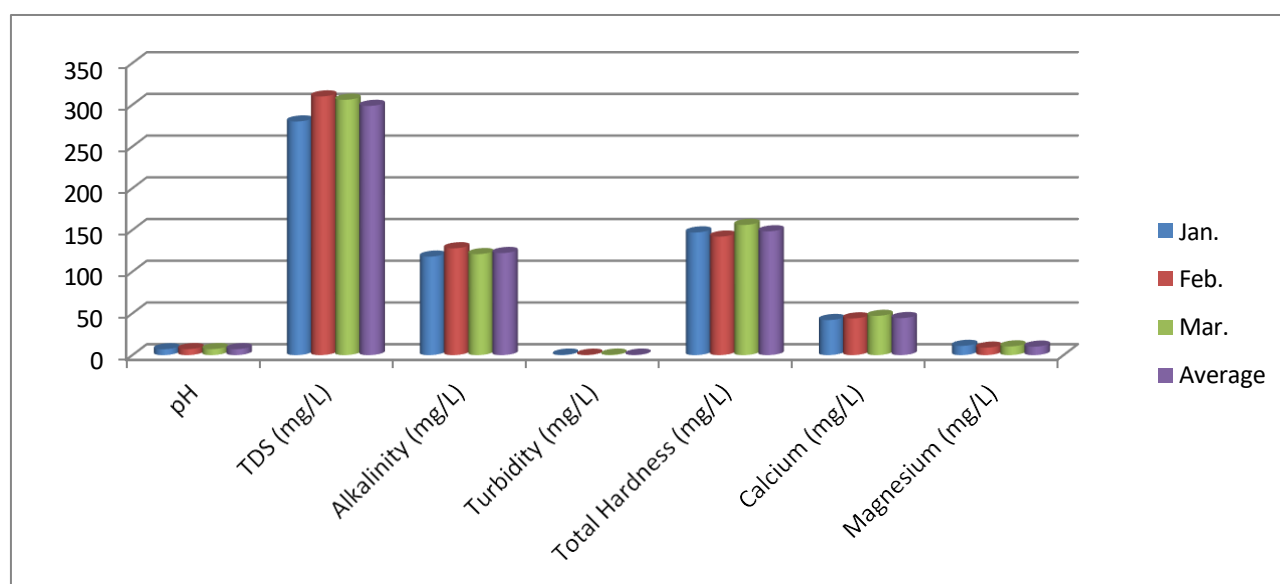
Calcium: The minimum Calcium was observed 42mg/L in January and maximum value was observed 47 mg/L in March. The Calcium is naturally present in water. It may dissolve from rocks such as limestone, marble, calcite, dolomite, gypsum, fluorite and apatite. Calcium is a determinant of water hardness, because it can be found in water as Ca^{2+} ions.

Magnesium: In the course of the present study the minimum Magnesium was determined 8.9 mg/L in February whereas 11 mg/L was maximum value of Magnesium in January. Magnesium has many different purposes and consequently may end up in water in many different ways. Chemical industries add magnesium to plastics and other materials as a fire protection measure or as filler. It also ends up in the environment from fertilizer application and from cattle feed. Magnesium Sulphate is applied in beer breweries, and magnesium hydroxide is applied as a flocculent in wastewater treatment plants.

Table:4. Physicochemical parameters analyzed at sampling site 3.

Parameters/Months	Jan.	Feb.	Mar.	Average	BIS Standard for DW (Permissible Limit)
pH	7.02	7.22	7.21	7.15	6.5-8.5
TDS (mg/L)	280	310	306	298.66	2000
Alkalinity(mg/L)	118	128	121	122.33	600
Turbidity(NTU)	0.5	0.4	0.5	0.46	5
Total Hardness(mg/L)	147	142	156	148.33	600
Calcium(mg/L)	42	44	47	44.33	200
Magnesium(mg/L)	11	8.9	10.4	10.1	100

Graph3: Showing the parameters at sampling sites 3.



Average of parameters of all sites

pH: In the course of the present study the minimum average of pH was observed 7.15 mg/L at site 3 whereas 7.91 mg/L was maximum value of pH at site 1 . As mentioned above this difference may be due to the fact that carbon-dioxide dissolved in water to form carbonic acid. Although this is a weak acid, large amount of it will lower the pH and when water with low pH values come in contact with certain chemicals and metals, this often makes them more poisonous than normal.

TDS: In the present study the minimum average of TDS was observed 272.33 mg/L at site 2 whereas 307.6 mg/L was maximum value of TDS at site 1. Most often, high levels of TDS are caused by the presence of potassium, chlorides and sodium. These ions have little or no short- term effects, but toxic ions (lead arsenic, cadmium, nitrate and others) may also be dissolved in the water.

Alkalinity: Throughout the present study the minimum average of Alkalinity was observed 107 mg/L at site 2 whereas 135.20 mg/L was maximum value of Alkalinity at site 1. It may be due to the presence of large amount of carbonates and bicarbonates ions from rocks and other resources.

Turbidity: In the course of the present study the minimum average of turbidity was observed 0.26 NTU at site 2 whereas 0.6 NTU was maximum value of turbidity at site 1. Turbidity in open water may be caused by the growth of phytoplankton. Human activities that disturb land, such as construction, mining and agriculture, can lead to high sediment levels entering water bodies during rainstorms due to storm water runoff.

Total Hardness: The minimum average of total hardness was observed 148.33mg/L at site 3. whereas 157.03 mg/L was maximum value of total hardness at site 1. This difference in hardness may arise due to the presence of Sulphate, carbonates, silicates arising from various industrial effluent and natural sources like rocks. Hardness is the property of water which prevents the lather formation with soap and increases the boiling points of water.

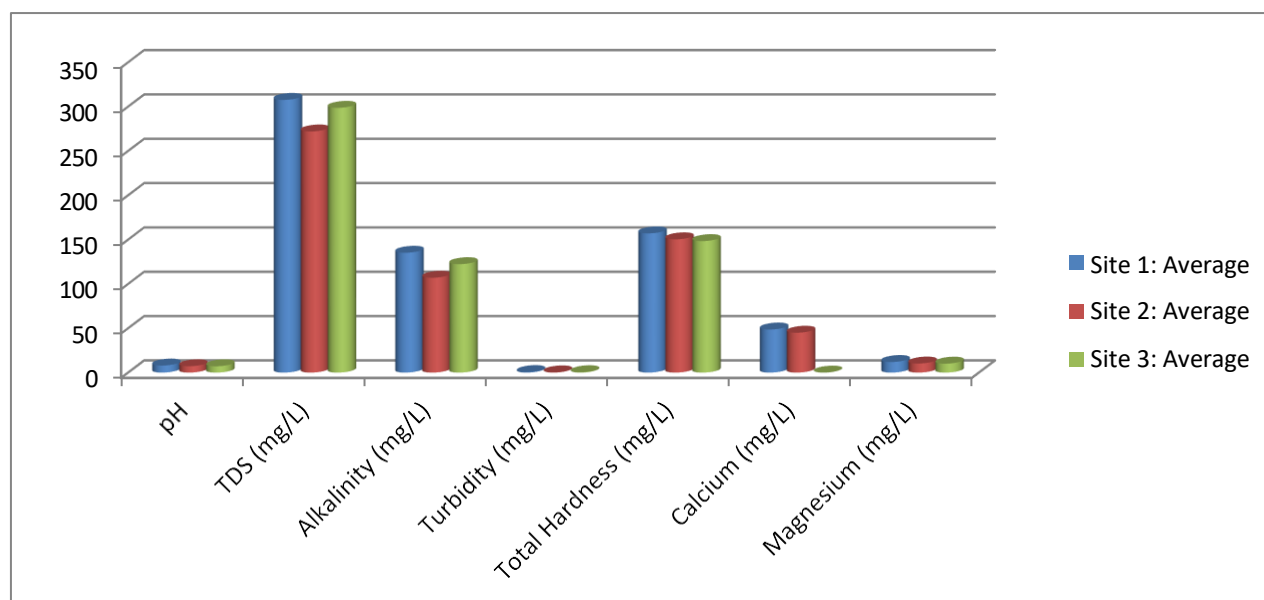
Calcium: In the present study the minimum average of Calcium was observed 44.33 mg/L at site 3 whereas 48.60 mg/L was maximum value of Calcium at site 1, The Calcium is naturally present in water. It may dissolve from rocks such as limestone, marble, calcite dolomite, gypsum, fluorite and apatite. Calcium is a determinant of water hardness, because it can be found in water as Ca^{2+} ions.

Magnesium: Throughout the present study the minimum average of Magnesium was observed 10.1 mg/L at site 3 whereas 11.96 mg/L was maximum value of Magnesium at site 1. Magnesium has many different purposes and consequently may end up in water in many different ways. Chemical industries add magnesium to plastics and other materials as a fire protection measure or as a filler. It also ends up in the environment from fertilizer application and from cattle feed. Magnesium Sulphate is applied in beer breweries, and magnesium hydroxide is applied as a flocculent in wastewater treatment plants.

Table5: Variation in average values of parameters reading of all the 3 Sites.

Sites/Parameters	Site1:Average	Site2:Average	Site3:Average
pH	7.98	7.22	7.16
TDS(mg/L)	307.8	274	298.67
Alkalinity(mg/L)	135.21	108	122.34
Turbidity(NTU)	0.7	0.27	0.47
Total Hardness(mg/L)	157.04	150.34	148.34
Calcium(mg/L)	48.61	46	44..34
Magnesium(mg/L)	11.97	10.24	10.2

Graph4: Showing the average value of all parameters at all sampling sites.



CHAPTER - 6

DISCUSSION

pH

pH measurements of various sites lead to conclude that the minimum pH is at site-3 and maximum at site-1. This may be due to the fact that carbon dioxide dissolves in water to form carbonic acid. Although this is a weak acid, large amounts of it will lower the pH and when waters with low pH values come into contact with certain chemicals and metals, this often makes them more poisonous than normal. Mandal *et al.*, (2012) observed more or less trend value has been recorded.

Hardness:

The hardness is caused primarily by the presence of cations such as Calcium and Magnesium and Anions such as Carbonates, Bicarbonates, and Chloride in water. Water Hardness has no known adverse effects. However, some evidence indicates its role in heart diseases and hardness of and above may cause kidney problems and stone formation. In the present study the maximum Hardness was observed at site-1 and the minimum was observed at site 3. Rao et al., (2013) observed the value of Total Hardness 221-474 (mg/l) in Vuyyuru, Part of East Coast of India.

Calcium:

Calcium is an important element to develop proper bone growth. Calcium content is very common in groundwater, because they are available in most of the rocks abundantly and also due to its higher solubility. The permissible limit of Calcium in groundwater is. In the present study the maximum Calcium was observed at site-1 and the minimum was observed at site-3. (Shayamala, R. et al., 2008) observed the value of calcium 49.48-81.76 (mg/l) in Vuyyuru, Part of East Coast of India.

Magnesium:

Magnesium usually occurs in lesser concentration than Calcium due to the fact that the dissolution of Magnesium rich mineral is slow process and that of calcium is more abundant in earth's crust. If the concentration of Magnesium in groundwater is more than the permissible limit, it causes unpleasant taste to the water. Magnesium is an essential ion for functioning of cells in enzyme activation, but at higher concentration, it is considered as laxative agent, whose deficiency may cause structural and functional changes in human beings. The permissible limit is 30 mg/l as per BIS standards. In the present study the maximum Magnesium was observed at site-1 and the minimum was observed at site 3. (Shayamala, R. et al., 2008) observed the value of

Magnesium 30-100 (mg/l) in Vuyyuru, Part of East Coast of India.

Alkalinity:

The total alkalinity values are ranging from 135.20, 10, 122.33 mg/l at site 1, site 2, site 3, respectively. The high alkalinity of groundwater may be due to the presence of bicarbonate and some salts. In the present study the maximum alkalinity was observed at site 1 and the minimum alkalinity was observed at site 2. Rao et al., (2013) observed the value of Alkalinity 308-488 (mg/l) in Vuyyuru, Part of East Coast of India.

Total Dissolved Solid

Maximum amount of T.D.S is determined at site-1 and minimum at site-3. Complete broke down solids (TDS) involve inorganic salts (mainly calcium, magnesium, potassium, sodium, bicarbonates, chlorides, and sulfates) and some little measures of natural issue that are disintegrated in water. It connotes the inorganic contamination heap of water framework. Vaithiyathan, T. and Sundaramoorthy, P. (2017) observed a more or less trend values has been recorded.

Turbidity

It is a significant factor that controls the vitality relationship at various tropic dimensions. It is basically an element of impression of light from the surface and is affected by the assimilation qualities of both water and of its disintegrated and particulate issue. Turbidity measurements show that the turbidity level of site-1 was found to be maximum and found minimum at site-3. Matta, G. (2014) observed a more or less trend value has been recorded.

CHAPTER - 7

SUMMARY

To estimate the ground water quality near Iol chemicals and pharmaceuticals area, physicochemical characteristics of water have been analyzed. Samples have been collected monthly covering three sites during the study period. A total of seven water quality parameters namely pH, total dissolved solids, alkalinity, turbidity, hardness, Ca^{2+} , Mg^{2+} , were analyzed for three sites by the following standard analytical methods of APHA. The results were compared with the recommended Standards for drinking water of BIS, to know the existing quality of water. While comparing the values of all water samples with respective recommended standards, it was observed that the values of the following parameters: pH, total dissolved solids, alkalinity, turbidity, hardness, Ca^{2+} , Mg^{2+} , were found within the desirable limit in all the samples during the study period. In all water tests, it was found that the some parameters are slightly high and essential water treatment is necessary for site-.1.

In summary, according to the results of this research, it is recommended that routine water testing be carried out to monitor the kind and extent of pollution. Individuals residing close to industrial areas face an increased likelihood of being exposed to groundwater pollution. This shows the importance of residents consuming treated water and experts supervising industrial wastewater discharge in the vicinity. Furthermore, companies need to set aside a small part of their earnings for effective waste management prior to disposal to guarantee a healthy and safe environment for individuals.

CHAPTER - 8

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